

Part 5 — Explainability and uncertainty

Complete tutorial for sciModelFlowR (version 1.0.2)

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1 Explaining without confusing importance with cause

A model can be right and still poorly understood. The practical question is twofold: **which variables matter** (importance) and **how the model responds** when a variable changes (profiles). `sciModelFlowR` separates the two and keeps the explanation tied to the design (no explaining on the test set).

```
field <- ler("field_yield")
v <- verdate$field_yield
no_out <- field[!field$plot_id %in% v$outlier_ids, ]

spec_p5 <- smf_experiment_spec(
  task = smf_task_spec("regression", "yield"),
  data = smf_data_spec("yield", c("nitrogen", "potassium", "water", "organic_matter",
                                   "block", "genotype"),
                        id_column = "plot_id", categorical = c("block", "genotype")),
  design = smf_design_spec(id_column = "plot_id"),
  preprocessing = smf_preprocess_spec("median"),
  resampling = smf_resampling_spec("holdout", train_prop = 0.7, seed = 21L),
  model = smf_model_spec("linear_regression", "stats"),
  metrics = list(smf_metric_spec("rmse")),
  reproducibility = smf_reproducibility_spec(21L)
)
fit_p5 <- smf_fit_experiment(spec_p5, no_out)
test_p5 <- no_out[fit_p5$split@test_index, ]

imp <- smf_permutation_importance(fit_p5, test_p5, n_repeats = 5L, seed = 9L)
tab_smf(imp[, c("feature", "importance", "sd", "n_repeats")],
        caption = "Permutation importance: RMSE drop when the variable is shuffled (5 repetitions)")
```

Table 1.1: Permutation importance: RMSE drop when the variable is shuffled (5 repetitions).

	feature	importance	sd	n_repeats
1	nitrogen	1080.581	108.764	5
3	water	169.790	51.689	5
2	potassium	52.276	14.571	5
6	genotype	13.414	2.537	5
4	organic_matter	9.077	7.469	5
5	block	-6.600	6.523	5

```
ggplot(imp, aes(stats::reorder(feature, importance), importance)) +
  geom_col(fill = cores_smf[["primaria"]], width = .6) +
  geom_errorbar(aes(ymin = importance - sd, ymax = importance + sd), width = .2) +
  coord_flip() +
  labs(x = NULL, y = "importance (RMSE drop, kg/ha)",
       title = "Permutation importance with SD") +
  ylim(min(0, min(imp$importance - imp$sd)), max(imp$importance + imp$sd) * 1.08)
```

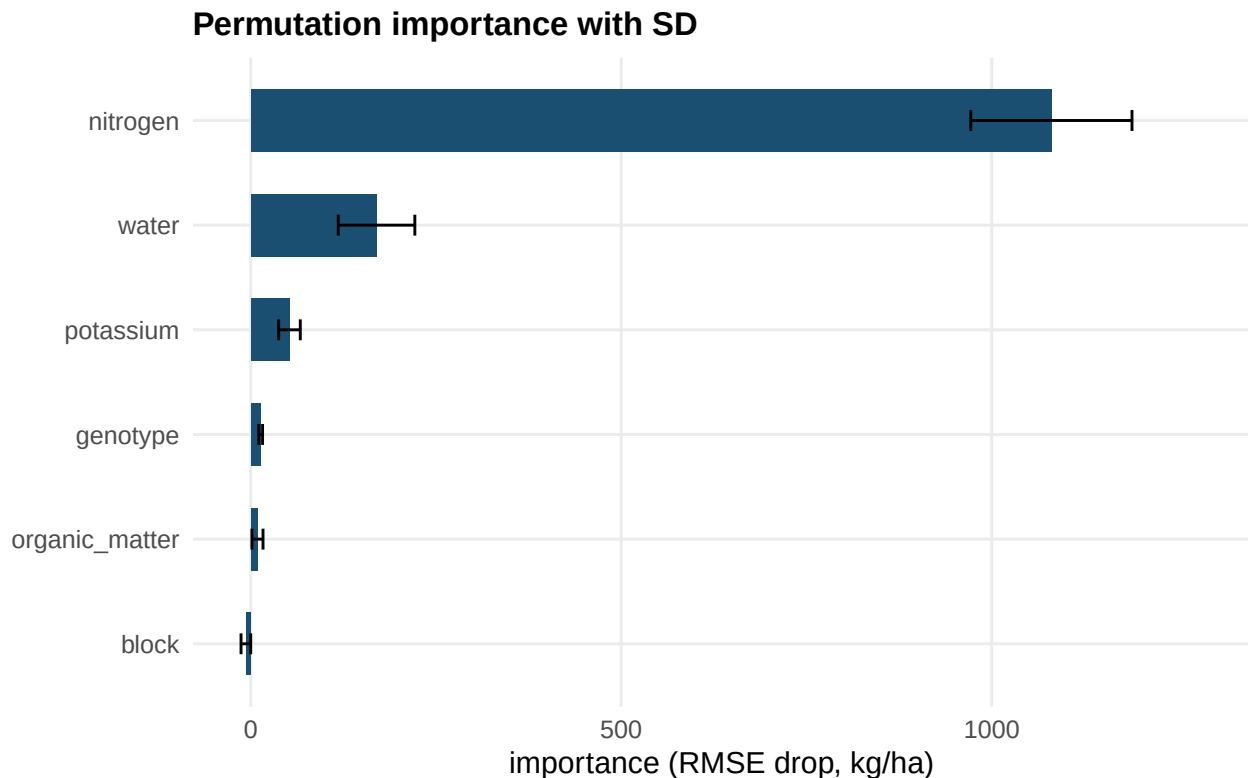


Figure 1.1: Importance with uncertainty: nitrogen dominates; water and potassium come far behind; block and genotype barely matter because they have few levels and moderate effects.

Profiles answer “how the model changes with the dose”. PDP, ICE and ALE answer the same question through different paths: PDP integrates over the marginal; ICE shows a curve per observation; ALE measures local effects in bins.

```
pdp_n <- smf_pdp(fit_p5, test_p5, features = "nitrogen", grid_size = 12L)
ale_n <- smf_ale(fit_p5, test_p5, features = "nitrogen", bins = 12L)
ice_n <- smf_ice(fit_p5, test_p5, features = "nitrogen", grid_size = 12L, n = 40L, seed = 7L)
cat("PDP rows:", nrow(pdp_n), "| ICE rows:", nrow(ice_n), "| ALE rows:", nrow(ale_n), "\n")
```

PDP rows: 12 | ICE rows: 480 | ALE rows: 12

```

p_pdp <- ggplot() +
  geom_line(data = ice_n, aes(value, prediction, group = row_id),
            colour = cores_smf[["claro"]], alpha = .4) +
  geom_line(data = pdp_n, aes(value, prediction), colour = cores_smf[["alerta"]],
            linewidth = 1.1) +
  labs(x = "nitrogen (kg/ha)", y = "predicted yield (kg/ha)",
        title = "PDP (red) over ICE (grey)")
p_ale <- ggplot(ale_n, aes(value, effect)) +
  geom_line(colour = cores_smf[["primaria"]], linewidth = 1.1) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(x = "nitrogen (kg/ha)", y = "accumulated local effect (kg/ha)",
        title = "ALE: local effect")
patchwork::wrap_plots(p_pdp, p_ale)

```

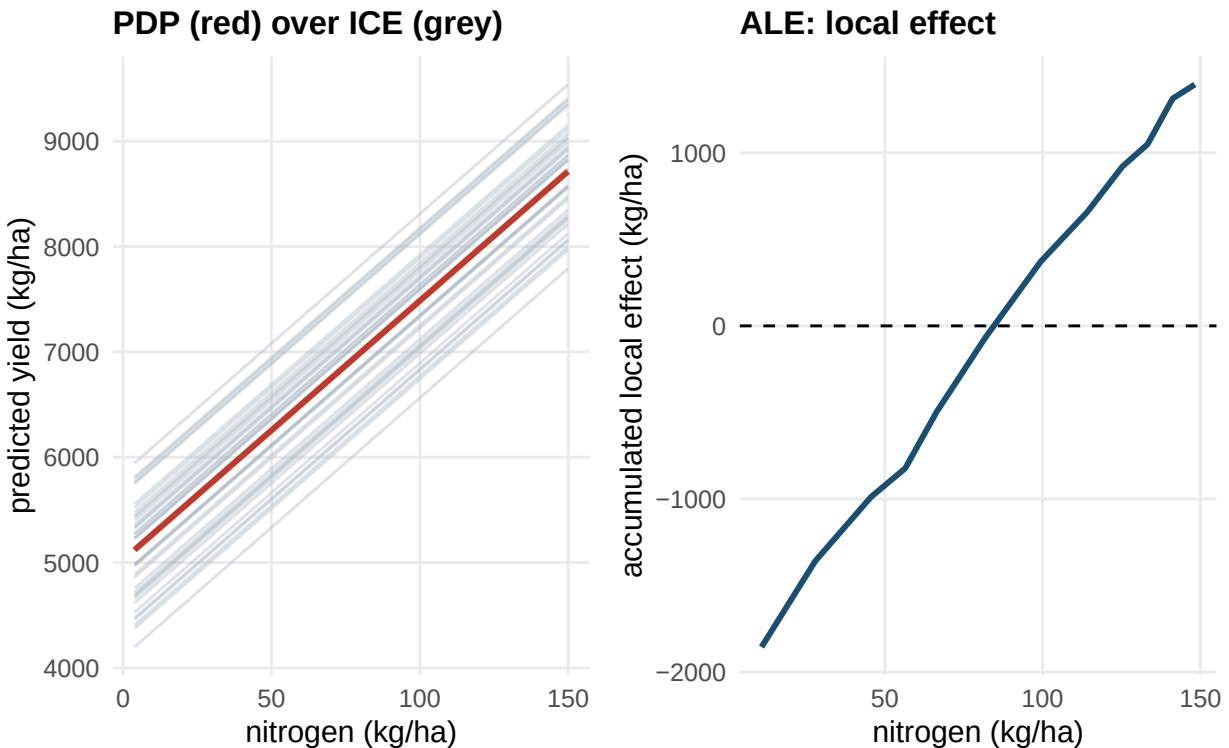


Figure 1.2: Dependence profiles for nitrogen: the mean PDP and per-plot ICE curves (left) and the accumulated local effect (right). The slope recovers ~ 25 kg/ha per unit.

```

loc <- smf_local_explain(fit_p5, test_p5[1, ], no_out, features = c("nitrogen", "water"))
tab_smf(loc, caption = "Local explanation of the first test plot: contribution of each variable")

```

Table 1.2: Local explanation of the first test plot: contribution of each variable to the difference between prediction and reference.

row_id	feature	original	reference	prediction	reference_prediction	contribution
1	nitrogen	146.8	87.7	8692.292	7237.908	1454.384
1	water	574.8	510	8692.292	8500.909	191.383

```
sh <- smf_shap(fit_p5, background = no_out[1:40, ], new_data = test_p5[1:4, ],
              nsim = 20L, seed = 3L)
tab_smf(sh[1:4, c("plot_id", "nitrogen", "water", "organic_matter")],
        caption = "SHAP values for four plots (additive, approximate attribution).")
```

Table 1.3: SHAP values for four plots (additive, approximate attribution).

plot_id	nitrogen	water	organic_matter
0	1466.345	395.061	37.264
0	-1219.080	77.930	80.612
0	-616.581	35.863	-40.935
0	218.776	-172.779	-6.470

i Interpretation

The three tools agree on direction and order: nitrogen dominates, the curve is almost straight with a slope close to the truth (25 kg/ha per unit) and the local explanation balances the books ($\text{contribution} = \text{prediction} - \text{reference}$). Importance **is not** cause: shuffling nitrogen also destroys the correlated structure of the trial; the reading is therefore “the model uses this variable”, not “this variable causes the effect”.

1.1 Explanation stability

An explanation from a single fit can be luck. `smf_explanation_stability()` repeats the explanation across folds and measures rank agreement.

```
spec_cv_p5 <- smf_experiment_spec(
  task = smf_task_spec("regression", "yield"),
  data = smf_data_spec("yield", c("nitrogen", "potassium", "water", "organic_matter",
                                   "block", "genotype"),
                      id_column = "plot_id", categorical = c("block", "genotype")),
  design = smf_design_spec(id_column = "plot_id"),
  preprocessing = smf_preprocess_spec("median"),
  resampling = smf_resampling_spec("kfold", n_splits = 4L, seed = 17L),
  model = smf_model_spec("linear_regression", "stats"),
  metrics = list(smf_metric_spec("rmse")),
```

```

    reproducibility = smf_reproducibility_spec(17L)
  )
  exspec <- smf_explain_spec("permutation", n_repeats = 3L, seed = 44L)
  xai_stability <- smf_explanation_stability(spec_cv_p5, no_out, exspec, top_k = 3L)
  tab_smf(xai_stability@summary, caption = "Mean importance across folds, with the frequency of

```

Table 1.4: Mean importance across folds, with the frequency of a positive signal.

feature	mean_importance	sd_importance	cv_importance	positive_fraction	n_folds
nitrogen	1146.098	90.310	0.079	1.0	4
water	180.266	37.930	0.210	1.0	4
potassium	100.424	15.385	0.153	1.0	4
block	-2.398	9.646	4.022	0.5	4
genotype	17.192	16.959	0.986	1.0	4
organic_matter	8.885	6.400	0.720	1.0	4

```

  tab_smf(xai_stability@rank_stability, caption = "Rank stability: mean position of each variable

```

Table 1.5: Rank stability: mean position of each variable across folds.

feature	mean_rank	sd_rank
nitrogen	1.0	0.000
water	2.0	0.000
potassium	3.0	0.000
block	5.5	1.000
genotype	4.5	0.577
organic_matter	5.0	0.816

! Interpretation

With 6 variables and 4 folds, the ranking is stable when **sd_rank** is small and **positive_fraction** is 1. Nitrogen appears at the top in every fold; weak-effect variables swap positions — and that is information, not a defect: unstable explanations must be communicated as unstable.

2 Uncertainty with a declared meaning

The package refuses to treat uncertainty types as synonyms. The semantics table is the contract: each interval promises something different, and none is interchangeable.

```
sem_u <- smf_uncertainty_semantics()
tab_smf(sem_u, caption = "Canonical semantics of interval types: what each promises and under
```

Table 2.1: Canonical semantics of interval types: what each promises and under which assumptions.

interval_type	target	probability_statement	interchangeable
confidence	parameter or estimand	repeated-sample procedure	FALSE
credible	parameter/latent quantity	posterior probability conditional on model/prior	FALSE
prediction	future response	sampling distribution conditional on fitted model	FALSE
posterior_predictive	future response	posterior predictive probability	FALSE
bootstrap	estimator/statistic	resampling approximation	FALSE
conformal	future response or class set	marginal coverage under conformal assumptions	FALSE

```
desc_conformal <- smf_uncertainty_descriptor("conformal", "future_response", "conformal",
                                             0.90, "exchangeability")
desc_bootstrap <- smf_uncertainty_descriptor("bootstrap", "estimator", "bootstrap",
                                             0.95, "resampling approximation")
val_con <- smf_uncertainty_validate(desc_conformal)
val_boot <- smf_uncertainty_validate(desc_bootstrap)
desc_tbl <- data.frame(
  label = c(smf_uncertainty_label(desc_conformal), smf_uncertainty_label(desc_bootstrap)),
  valid = c(val_con$valid, val_boot$valid)
)
tab_smf(desc_tbl, caption = "Uncertainty labels: source, target and assumption are part of th
```

Table 2.2: Uncertainty labels: source, target and assumption are part of the name.

label	valid
conformal 90% [conformal; target=future_response]	TRUE
bootstrap 95% [bootstrap; target=estimator]	TRUE

label	valid
-------	-------

```
decomp <- smf_uncertainty_decompose(aleatoric = 0.4, epistemic = 0.6, total = 1.0,
                                   method = "ensemble", identified = TRUE)
print(decomp)
```

```
<sciModelFlowR UncertaintyDecomposition>
method: ensemble
identified: TRUE
```

```
refusal_decomp <- tryCatch(
  smf_uncertainty_decompose(aleatoric = 0.4, epistemic = 0.6, method = "unknown",
                           identified = FALSE),
  error = function(e) conditionMessage(e))
cat("decomposition without identification:", refusal_decomp, "\n")
```

decomposition without identification: Aleatoric/epistemic decomposition was requested without a

```
refusal_dist <- tryCatch(
  smf_uncertainty_from_distribution(
    smf_prediction_distribution("parametric",
                               list(distribution = "normal", mean = 5000, sd = 300))),
  error = function(e) conditionMessage(e))
cat("distribution without identified decomposition:", refusal_dist, "\n")
```

distribution without identified decomposition: This prediction distribution does not expose an

Trap

The message above is a deliberate refusal: the package **does not fabricate** an aleatoric/epistemic decomposition that the object cannot support. Without a procedure that separates the sources (ensemble, replicates, hierarchy), the number would be fiction with a technical appearance.

3 Conformal: promised coverage, measured coverage

The conformal guarantee is **marginal and under exchangeability** — in the heteroscedastic trial, that is exactly the kind of promise that must be measured. The split below freezes three roles: training, calibration and untouched test.

```
complete <- no_out[stats::complete.cases(
  no_out[, c("nitrogen", "potassium", "water", "organic_matter")]), ]
set.seed(20260919)
order_idx <- sample.int(nrow(complete))
i_tr <- order_idx[1:90]; i_cal <- order_idx[91:120]; i_te <- order_idx[121:149]
f_con <- stats::lm(yield ~ nitrogen + potassium + water + organic_matter,
  data = complete[i_tr, ])
p_cal <- unname(stats::predict(f_con, complete[i_cal, ]))
p_te <- unname(stats::predict(f_con, complete[i_te, ]))
con <- smf_conformal_fit(truth = complete$yield[i_cal], prediction = p_cal,
  spec = smf_conformal_spec("split", level = 0.90),
  design = smf_design_spec(id_column = "plot_id"),
  calibration_ids = complete$plot_id[i_cal],
  final_test_ids = complete$plot_id[i_te])
inter <- smf_conformal_predict(con, prediction = p_te)
coverage <- smf_conformal_coverage(complete$yield[i_te], inter)
tab_smf(coverage, caption = "Empirical coverage of the 90% conformal interval on the untouched test set.")
```

Table 3.1: Empirical coverage of the 90% conformal interval on the untouched test set.

n	coverage	mean_width	median_width
29	0.897	837.043	837.043

```
diag_con <- smf_conformal_diagnose(con, truth = complete$yield[i_te], prediction = inter)
tab_smf(diag_con$summary, caption = "Conformal diagnostics: coverage against the target, with declared tolerance.")
```

Table 3.2: Conformal diagnostics: coverage against the target, with declared tolerance.

n	coverage	mean_width	median_width
29	0.897	837.043	837.043

```

tab_smf(data.frame(target = diag_con$target, observed = diag_con$observed,
                    monte_carlo_se = diag_con$monte_carlo_se,
                    tolerance = diag_con$tolerance),
        caption = "Target, observed coverage and the sampling margin of error of coverage.")

```

Table 3.3: Target, observed coverage and the sampling margin of error of coverage.

target	observed	monte_carlo_se	tolerance
0.9	0.897	0.056	0.111

```

ord <- order(p_te)
con_df <- data.frame(plot = seq_along(p_te), pred = p_te[ord],
                    lo = inter$.pred_lower[ord], hi = inter$.pred_upper[ord],
                    obs = complete$yield[i_te][ord])
ggplot(con_df, aes(plot, pred)) +
  geom_ribbon(aes(ymin = lo, ymax = hi), fill = cores_smf[["claro"]], alpha = .5) +
  geom_point(aes(y = obs), size = 1.6, colour = cores_smf[["alerta"]]) +
  geom_line(colour = cores_smf[["primaria"]]) +
  labs(x = "test plot (ordered by prediction)", y = "yield (kg/ha)",
       title = "Split conformal intervals at 90%",
       subtitle = paste0("Measured coverage: ", fmt(100 * coverage$coverage, 1),
                        "% | mean width: ", fmt(coverage$mean_width, 0), " kg/ha"))

```

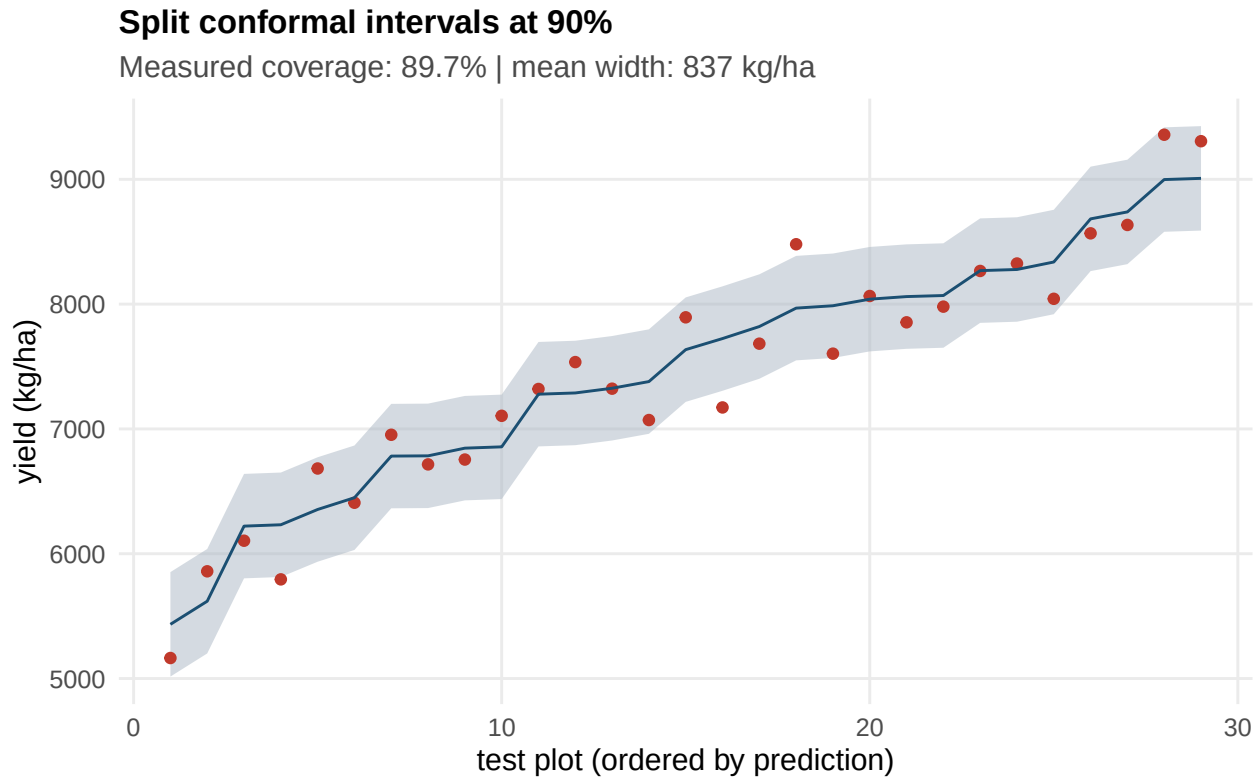


Figure 3.1: Conformal intervals ordered by prediction: width is constant (split conformal), and coverage is counted on the test set, not presumed.

In classification, the target is not an interval but a **set** of labels with guaranteed coverage (APS). The set size is the uncertainty measure.

```
disease <- ler("disease_binary")
disease$disease <- factor(disease$disease, levels = c(0, 1))
set.seed(20260919)
od <- sample.int(nrow(disease))
tr_c <- od[1:250]; cal_c <- od[251:350]; te_c <- od[351:400]
glm_c <- stats::glm(disease ~ x1 + x2 + x3 + x4, data = disease[tr_c, ],
  family = stats::binomial())
P_cal <- cbind(`0` = 1 - stats::predict(glm_c, disease[cal_c, ], type = "response"),
  `1` = stats::predict(glm_c, disease[cal_c, ], type = "response"))
P_te_c <- cbind(`0` = 1 - stats::predict(glm_c, disease[te_c, ], type = "response"),
  `1` = stats::predict(glm_c, disease[te_c, ], type = "response"))
aps <- smf_conformal_fit(disease$disease[cal_c], probabilities = P_cal,
  spec = smf_conformal_spec("aps", level = 0.90))
sets <- smf_conformal_predict(aps, probabilities = P_te_c)
cov_cls <- smf_conformal_coverage(disease$disease[te_c], sets)
tab_smf(cov_cls, caption = "APS conformal sets: coverage and mean set size on the test set.")
```

Table 3.4: APS conformal sets: coverage and mean set size on the test set.

n	coverage	mean_set_size	median_set_size
50	1	2	2

i Interpretation

Coverage of 90% is measured, not presumed: with 29 test plots, coverage was 89.7% and mean width 837 kg/ha — constant, because split conformal uses a single quantile of the scores. In classification, coverage is 100.0% with an average set of 2.00 labels: the more uncertain the case, the larger the set.

4 Transport: does the model work outside the domain?

Block B4 has its own effect and is a natural “external domain” candidate. External validation uses the **already fitted** model, with no refit and no recalibration.

```
domain_df <- no_out
domain_df$domain <- ifelse(domain_df$block == "B4", "external", "development")
spec_dev <- smf_experiment_spec(
  task = smf_task_spec("regression", "yield"),
  data = smf_data_spec("yield", c("nitrogen", "potassium", "water", "organic_matter"),
    id_column = "plot_id"),
  design = smf_design_spec(id_column = "plot_id"),
  preprocessing = smf_preprocess_spec("median"),
  resampling = smf_resampling_spec("holdout", train_prop = 0.8, seed = 31L),
  model = smf_model_spec("linear_regression", "stats"),
  metrics = list(smf_metric_spec("rmse"), smf_metric_spec("mae")),
  reproducibility = smf_reproducibility_spec(31L)
)
dev <- domain_df[domain_df$domain == "development", ]
ext <- domain_df[domain_df$domain == "external", ]
fit_dev <- smf_fit_experiment(spec_dev, dev)
val_ext <- smf_external_validate(fit_dev, ext, domain_name = "external")
cmp_ext <- smf_compare_validation_domains(fit_dev, val_ext)
tab_smf(cmp_ext$comparison, caption = "Internal and external metrics side by side: the same m
```

Table 4.1: Internal and external metrics side by side: the same model, two domains.

metric	internal	external	delta	direction
rmse	366.263	416.144	49.881	minimize
mae	258.757	279.990	21.234	minimize

```
ext_df <- data.frame(
  domain = c("internal (holdout)", "external (block B4)"),
  rmse = c(cmp_ext$internal$value[cmp_ext$internal$metric == "rmse"][1],
    cmp_ext$external$value[cmp_ext$external$metric == "rmse"][1])
)
ggplot(ext_df, aes(domain, rmse, fill = domain)) +
  geom_col(width = .5, show.legend = FALSE) +
  geom_text(aes(label = fmt(rmse, 0)), vjust = -0.4) +
```

```
scale_fill_manual(values = c(cores_smf[["primaria"]], cores_smf[["alerta"]])) +
labs(x = NULL, y = "RMSE (kg/ha)",
     title = "Internal vs external validation") +
ylim(0, max(ext_df$rmse) * 1.12)
```

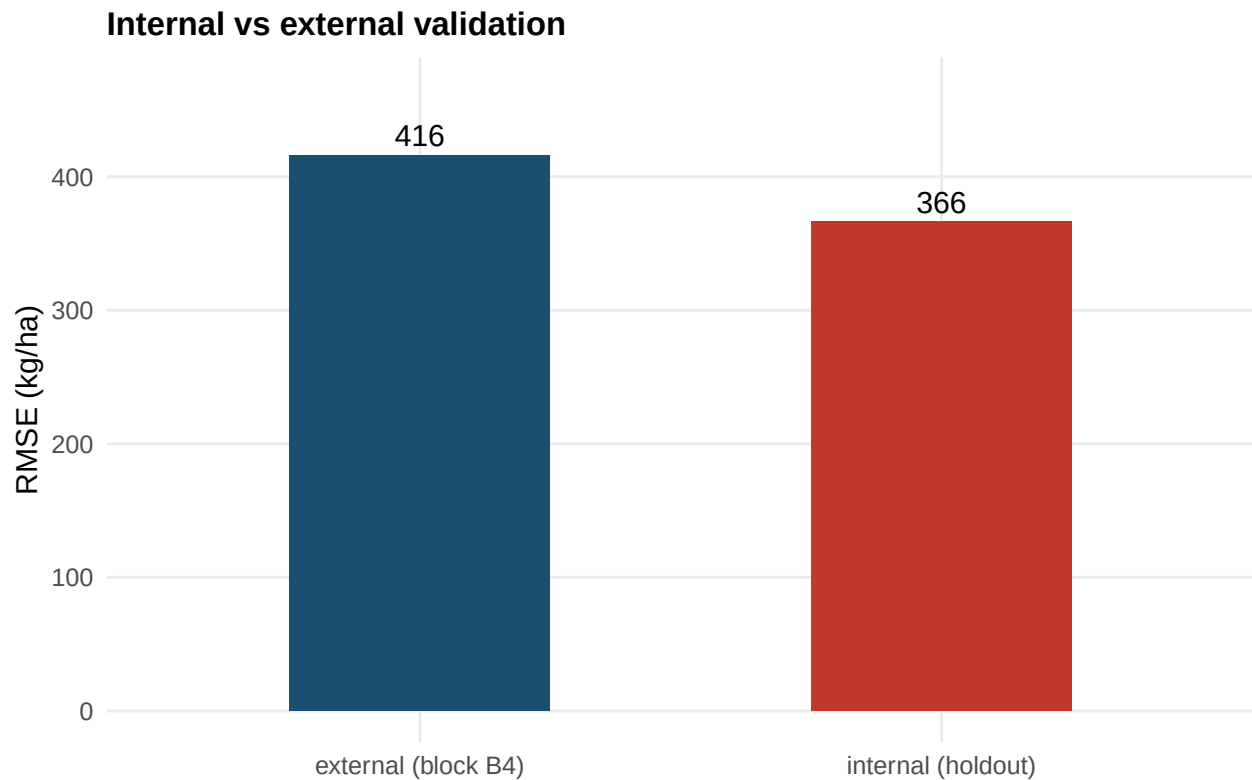


Figure 4.1: Transport degradation: block B4 has a +60 kg/ha effect and the model fitted on development does not know it — the difference shows up in external validation.

! Interpretation

External RMSE is 416 against 366 internally — the difference is the price of transporting a model to a block it never saw. No refit was done: the package measures the transport of the frozen artefact, which is what happens in practice.

5 Part 5 exercises

Exercise 5.1 — Importance versus causality

Nitrogen importance is huge, but water and potassium also enter the simulated truth. Propose an experiment (not computational) that would separate causal effect from association in this trial.

Exercise 5.2 — PDP versus ALE

Compare the PDP slope with the ALE slope for **nitrogen**. Do both match the truth (25)? In which situation would they diverge?

Exercise 5.3 — Conformal and heteroscedasticity

Measure conformal coverage **by nitrogen band** (low versus high doses). Does the 90% marginal coverage hold within each band? What does this say about the marginal guarantee?

Exercise 5.4 — Set size as uncertainty

In APS, order cases by set size. Does size track the predicted probability of the majority class? Build a figure.

Solutions

5.1. Randomisation: nitrogen doses drawn independently of the other variables (as in the simulation). In observational data, association does not identify the effect; an explicit causal design would be needed.

5.2. PDP and ALE coincide here because the model is linear and the variables are independent. With strong interactions or correlated predictors, PDP integrates over unlikely combinations and ALE localises the effect — the curves diverge.

5.3. Marginal 90% coverage does not imply 90% within each band: under heteroscedasticity, high doses tend to fall below and low doses above. That is why Part 6 presents **local** conformal (CQR) and per-band reading.

5.4. Yes: the set grows when the maximum probability is low, that is, when the model is unsure. The expected figure is a negative relationship between the predicted class probability and the set size.